

THE UNIVERSITY OF FAISALABAD

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Air Quality Prediction Using Machine Learning and Data Mining Techniques

A Data Mining Project Report

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Certificate

This is to certify that the project report titled "**Air Quality Prediction Using Machine Learning and Data Mining Techniques**" has been successfully carried out and submitted by **Abdul Manan Tanveer (Reg No. 2023-BS-AI-173)**, **Haziq Mustafa (Reg No. 2023-BS-AI-181)**, **Daniyal Ahmad (Reg No. 2023-BS-AI-183)**, and **Hassan Shahbaz (Reg No. 2024-BS-AI-159)**, students of BS Artificial Intelligence, in partial fulfilment of the requirements of the Data Mining course.

The work presented in this report is the result of the students' own effort. All experiments, source code, and results contained herein were prepared and verified by the students under the supervision of the course instructor.

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Abstract

Air pollution poses an increasingly serious threat to public health and urban ecosystems, making the accurate prediction of atmospheric pollutant levels an essential data-driven task. This project applies advanced machine learning and data mining techniques to predict the concentration of carbon monoxide (**CO(GT)**) in ambient air utilizing the well-known UCI Air Quality dataset. The dataset consists of 9,357 hourly readings collected over the span of a full year (March 2004 – February 2005) from a multi-sensor chemical device deployed in a heavily polluted Italian city.

The raw dataset was cleaned using systematic preprocessing procedures: handling the unique missing-value placeholder code (-200), omitting the NMHC(GT) attribute due to an excessive proportion of missing observations, and imputing any remaining sequence gaps using robust column-wise median values. Following rigorous exploratory data analysis and feature selection, three baseline and ensemble regression models—Linear Regression, Decision Tree, and Random Forest Regressor—were successfully trained on 7,485 structural observations and evaluated on 1,872 unseen test samples.

The Random Forest ensemble model achieved optimal performance, yielding an R^2 score of 0.905, a Mean Absolute Error (MAE) of 0.259, and a Root Mean Squared Error (RMSE) of 0.421. These empirical results confidently substantiate that ensemble tree-based methodologies are highly reliable for localized air-quality forecasting applications and can robustly enhance low-cost, multi-sensor environmental monitoring networks.

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1. Introduction

Air quality degradation represents one of the most pressing environmental and epidemiological crises of the modern industrial era. Rapid global urbanization, dense vehicular traffic, and sustained industrial manufacturing emit substantial volumes of dangerous chemical pollutants, including carbon monoxide (CO), nitrogen oxides (NO_x , NO_2), and aromatic hydrocarbons such as benzene (C_6H_6) directly into the troposphere. Sustained human exposure to elevated concentrations of these gaseous pollutants is fundamentally correlated with critical respiratory conditions, immune degradation, and chronic cardiovascular diseases.

Consequently, predicting local pollutant spikes well in advance provides municipalities, environmental regulatory bodies, and public citizens with actionable notice to implement critical preventive measures. Traditional chemical air-monitoring reference stations deliver extreme analytical accuracy, but their deployment is restricted by massive capital costs and high infrastructure density requirements. Compact, low-cost solid-state metal-oxide electronic sensors offer a viable method to scale geographical coverage, but their crude output signals exhibit temporal drift and environmental cross-sensitivity that demand rigorous post-calibration models.

Data mining frameworks and machine learning algorithms supply the perfect analytical apparatus: they automatically interpret complex multidimensional mapping equations between cheap hardware responses, ambient meteorological data, and the true underlying chemical concentration. This project maps and builds a responsive predictive environment utilizing the comprehensive UCI Air Quality dataset.

2. Problem Statement

High-fidelity, actionable air-quality information remains geographically sparse because standard chemical reference monitoring facilities are capital-intensive and rigidly bounded in number. Low-cost electrochemical and metal-oxide sensors generate multi-axis electrical responses that do not translate directly to target pollutant mass concentrations without advanced computation, and their transient signal metrics are heavily distorted by seasonal fluctuations in ambient temperature and relative humidity. The primary problem is to design an optimized, resilient machine learning model capable of accurately interpolating the true ground-truth carbon-monoxide ($\text{CO}_{(GT)}$) level purely from a set of low-cost cross-sensitive sensor responses and basic weather parameters, enabling municipal air quality monitoring networks to scale cost-effectively.

3. Objectives

- To extract, map, and conceptually model a multi-sensor real-world air-quality repository.
- To standardize and clean the underlying data using robust missing-value filtering and median statistical imputations.
- To execute exploratory data analysis (EDA) to map structural distributions and daily cyclic variations.
- To configure, optimize, and train multiple baseline and non-linear machine learning regression topologies.
- To benchmark mathematical model variations using R^2 , MAE, and RMSE metrics to lock the optimal architecture.
- To demonstrate high-fidelity prediction of **CO** gas concentrations across unseen environmental conditions.

4. Literature Review

The integration of statistical machine learning algorithms for atmospheric pollutant forecasting is an active domain within computing and environmental sciences. Aslan (2025), writing in *Water, Air, & Soil Pollution*, investigated the identical 9,357 hourly UCI sensor data profile and benchmarked ten separate regression techniques; their findings confirmed that ensemble tree methods and hyperparameter-optimized regressors yielded the highest overall prediction accuracies, directly substantiating the model design strategies executed in this project.

Méndez, Merayo, and Núñez (2023) authored an extensive survey in *Artificial Intelligence Review* spanning over one hundred foundational investigations, determining that ensemble tree-based architectures and sequence-aware time-series networks remain overwhelmingly successful for monitoring major particulate and gaseous pollutants like $PM_{2.5}$ and NO_2 , while concurrently emphasizing the crucial need for transparent model interpretations. Further, a 2024 experimental brief published in *Atmosphere* approached target air index tracking through multiclass classification setups, illustrating that targeted feature-space selection enhances validation performance. Houdou et al. (2024) explored explainable AI (XAI) frameworks, highlighting that tools such as SHAP value calculation simplify complex multi-layer model predictions for public policymakers.

The seminal dataset introductory paper by De Vito et al. (2008) pioneered this multi-sensor hardware setup, showing that on-field calibration of metal-oxide films via co-located reference stations could accurately track aromatic compound paths. Together, this rich matrix of literature establishes that our design—cleaning raw sensor feeds, conducting temporal feature processing, and comparing non-linear ensemble tree regressors—is well grounded in contemporary research.

5. Methodology

This implementation conforms to the standard CRISP-DM data mining lifecycle:

1. **Data Collection:** Programmatic extraction of the structured UCI Air Quality multi-sensor record matrix.
2. **Data Cleaning:** Universal conversion of the -200 noise flag into standard nulls, full exclusion of the corrupted NMHC(GT) parameter (~90% missing fraction), and executing row-wise target restorations via feature medians.
3. **Feature Engineering:** Splitting raw timestamps to derive independent *Hour* and *Month* coordinates, mapping diurnal and cyclic variations.
4. **Exploratory Analysis (EDA):** Extracting descriptive statistics, plotting probability distributions, and compiling correlation heatmaps.

5. **Feature Space Construction:** Retaining high-value sensor parameters, peripheral gas tracking metrics, and localized weather factors, setting true *CO(GT)* as the target vector.
6. **Model Construction:** Partitioning data into an 80% training block and a 20% validation split, executing z-score standard scaling, and fitting Linear Regression, Decision Tree, and Random Forest architectures.
7. **Evaluation & Selection:** Scoring performance across multiple statistical indexes to isolate the most reliable deployment architecture.

6. System Requirements

Hardware Specifications

- **Processor:** x86 or x64 Dual-Core CPU (Intel Core i3/Ryzen 3 equivalent or superior).
- **Memory:** 4 GB system RAM minimum (8 GB or greater highly recommended for large tree ensembles).
- **Storage:** Minimum 1 GB of available structural disk storage for software environments and local repository cache.

Software Configurations

- **Operating System:** Microsoft Windows 10/11, Apple macOS (Big Sur or newer), or mainstream Linux distributions (Ubuntu/Fedora).
- **Core Runtime:** Python 3.10 or newer local interpreter.
- **Development Interface:** Jupyter Notebook ecosystem managed via Anaconda, or Microsoft Visual Studio Code.

7. Tools & Technologies

Tool / Library	Analytical Purpose & Functional Scope
Python 3	Primary high-level programming language and core computational engine.
Jupyter Notebook	Interactive development environment for step-by-step model execution and documentation.
Pandas	High-performance relational data structure manipulation, parsing, and dataframe alignment.
NumPy	Multidimensional vectorized matrix mathematics and scientific computing routines.
Matplotlib & Seaborn	Static and statistical data visualization suites for building probability and trend graphs.

Tool / Library	Analytical Purpose & Functional Scope
Scikit-learn	Industry-standard machine learning library for model training, scaling, and evaluation metrics.

8. Dataset Details

The primary experimental repository contains 9,357 distinct hourly records generated by an advanced chemical multi-sensor device deployed in a highly congested Italian urban zone from March 2004 to February 2005. Each observation consists of validated ground-truth chemical readouts alongside parallel signals from five distinct co-located metal-oxide sensor elements and high-precision physical weather metrics. System nulls or signal dropouts are uniformly represented by a static -200 marker.

Column Identifier	Feature Meaning & Measured Parameter
CO(GT)	True carbon-monoxide mass concentration (mg/m ³), serving as our primary prediction target.
C6H6(GT), NOx(GT), NO2(GT)	True ground-truth chemical concentrations for Benzene (µg/m ³), Nitrogen Oxides (ppb), and Nitrogen Dioxide (mg/m ³).
PT08.S1 ... PT08.S5	Raw hourly average voltage/resistance responses from 5 custom-tuned metal-oxide gas sensor films (CO, NMHC, C6H6, NOx, NO2).
T, RH, AH	Ambient Temperature (°C), Relative Humidity (%), and absolute thermodynamic water content (Absolute Humidity).

Data Origin: UCI Machine Learning Repository (archive.ics.uci.edu/ml/datasets/Air+Quality)

9. Implementation

The comprehensive python codebase is structured within the attached interactive workspace notebook: `Air_Quality_Prediction.ipynb`. Preprocessing pipelines successfully converted numerical -200 flags into standard Python nulls, pruned the highly unpopulated NMHC(GT) feature vector, and filled localized attribute gaps via median column metrics. Time attributes (Hour, Month) were mapped to extract seasonal profiles.

10. Results and Discussion

Following training on 7,485 training data points, the predictive systems were benchmarked against 1,872 independent evaluation profiles. Detailed scoring across three critical indices is recorded in the table below (higher R^2 indicates stronger variance modeling; lower MAE/RMSE signify minor residuals):

Evaluated Model Architecture	R ² Score	MAE	RMSE
Random Forest Regressor	0.905	0.259	0.421
Linear Regression	0.834	0.379	0.557
Decision Tree Regressor	0.815	0.346	0.589

The Random Forest ensemble clearly outmatched other architectures, explaining approximately 90.5% of the total mathematical variation in the true carbon-monoxide target vector. This outcome confirms the prevailing literature that non-linear multi-tree aggregation models isolate intricate multi-variable dependencies much better than plain linear parameters.

Random Forest Relative Feature Importance

Feature Name	Relative Mathematical Importance
PT08.S2(NMHC) [Sensor Target 2]	0.529
NOx(GT) [True Nitrogen Oxides]	0.196
C6H6(GT) [True Benzene Level]	0.118
PT08.S1(CO) [Sensor Target 1]	0.028

Feature Name	Relative Mathematical Importance
NO2(GT) [True Nitrogen Dioxide]	0.025
Hour [Diurnal Cyclic Feature]	0.023

11. Conclusion

This project successfully constructed and deployed an end-to-end data mining and machine learning framework designed to transform raw cross-sensitive electronic sensor inputs into highly accurate ambient carbon-monoxide readings. Following data pipeline implementation and EDA over a comprehensive annual collection of 9,357 instances, three regression algorithms were systematically benchmarked. The Random Forest ensemble yielded superior accuracy ($R^2 = 0.905$), demonstrating that advanced data mining techniques can successfully calibrate and extract high-value analytics from low-cost environmental sensor arrays. This work outlines a practical, highly reproducible deployment pattern for scaling municipal environmental monitoring networks.

12. Future Scope

- **Sequence Deep Learning:** Transitioning to recurrent neural network architectures (such as LSTM or GRU) to model long-term seasonal and daily atmospheric trends.
- **Categorical AQI Classification:** Designing parallel classification heads to predict standard Air Quality Index categories (Good, Moderate, Unhealthy).
- **Hyperparameter Optimization:** Applying automated grid searches and cross-validation pipelines to squeeze out marginal error metrics.
- **Explainable AI Deployment:** Linking local SHAP and LIME explanatory layers to visualize feature impacts interactively for city management staff.
- **Web App Integration:** Packaging core serialization model binaries into web framework APIs (e.g., FastAPI/Streamlit) for real-time live reporting.

13. References

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